



EMPIRISYS LTD

The Empirisys Competitive Intelligence Engine (CIE)

A Sustained Competitive Advantage in the
HSE Software Market

PROGRAMME

MBA Accelerator Project ·
Module BST906

STUDENT

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01

Executive Summary

How a working prototype turns a real differentiator into a defensible, saleable advantage.

1. Executive Summary

Empirisys sells safety intelligence to high-hazard industries, where the buyer's decision is not about convenience but about preventing fatalities and managing the personal and criminal liability its Health, Safety and Environment (HSE) leaders carry by law. Its platform, BOOST, is a *System of Intelligence* : as Product Manager Andrea Rossi explained, an AI engine that reads free-text safety data and prioritises where risk is building, rather than a *System of Record* (Enablon, Sphera, DNV Synergi Life, Cority) that logs incidents but does not tell an operator what to do next. This engagement addressed the commercial problem defined by Head of Sales Joe Alner: despite a genuinely better product, Empirisys loses winnable deals to incumbent inertia and to a “parity illusion” in which every vendor now claims “AI,” leaving liable buyers unable to tell real safety intelligence from marketing language.

The response was a working full-stack prototype, The Empirisys Competitive Intelligence Engine, not a slide deck. Built with the architecture agreed with engineers David Riskus and Peter, it is a Next.js application whose dashboards track competitors on the dimensions Joe Alner sells against (HSE focus, UK presence and client overlap), and whose governed AI console lets the sales team query that landscape in natural language and receive on-message counter-positioning on demand. The AI is not left to chance: server-controlled prompts encode Empirisys's doctrine directly into the model, so it consistently frames incumbents as data-capture tools and generic enterprise AI as lacking the repeatability, auditability and safety governance high-hazard sites demand.

A mid-build pivot sharpened the build: generic CRM and lead-generation modules were cut once client feedback showed that a competitor-analysis tool becomes weaker, not stronger, when diluted with commodity features. Every screen was held to one test, “Does this help Empirisys understand or beat a competitor?”, trading breadth for a defensible product identity.

The analysis that follows shows this identity is commercially durable. Porter's Five Forces finds an industry of intense rivalry and a rising substitution threat from general-purpose AI, but one insulated by high switching costs and regulatory complexity (notably COMAH) that reward a specialist. VRIO confirms that BOOST's domain-trained language model and Empirisys's embedded advisory expertise are Valuable, Rare, costly to Imitate and Organised for: the conditions of a *sustained competitive advantage* . Rivals' retrofitted AI, by contrast, registers only as parity.

The recommendations are prioritised on a Value versus Risk/Cost matrix. The headline recommendation is to deploy the Intelligence Engine to Joe Alner's sales team now (a high-value, low-cost move this prototype already proves), while deferring capital-intensive work such as autonomous live data-scraping to a later, de-risked phase. The upshot is a Hub that converts an abstract differentiator into demonstrable proof and gives leadership an ownable competitive position.

02

Introduction: Context & Problem Statement

A liability-driven market, a category in transition, and a superior product that does not sell itself.

2.1 The Operating Environment

Empirisys serves high-hazard sectors (oil and gas, offshore energy, chemical processing and heavy manufacturing) where a system failure is measured not in downtime but in loss of life, environmental catastrophe and regulatory sanction. Under the Health and Safety at Work etc. Act 1974 and the Control of Major Accident Hazards (COMAH) Regulations 2015, individual directors and senior managers can face personal liability, prosecution and disqualification where safety failures stem from their neglect. The buyers Empirisys must persuade, senior HSE Directors and operational leaders, are therefore not procuring a productivity tool but a defence against the worst day of their professional life. That context is not incidental to the strategy; it is the strategy.

This is the commercial lever the project is built on. A vendor seen merely to help record what happened competes on features and price; a vendor that can credibly help *prevent* the incident, and evidence that prevention to a board and a regulator, competes on something liable leaders will pay a premium for and defend internally.

2.2 The Category Shift: From Systems of Record to Systems of Intelligence

Enterprise software is migrating from *Systems of Record*, which store and report structured data, toward *Systems of Intelligence*, which interpret unstructured information and recommend action (Moore, 2011); safety software has been slower than most. The HSE incumbents (Enablon, Sphera, DNV's Synergi Life and Cority) were built as Systems of Record. They log incidents, near-misses and audits well, but were not architected to read the narrative of a safety report, connect weak signals across sites, or tell a manager which of a thousand data points warrants action today. Their answer to the AI era has largely been to bolt predictive "layers" onto rigid, monolithic architectures, an approach this report shows is strategically fragile.

Empirisys sits on the other side of this divide by design. BOOST is a real decision-support layer, and its companion SENSE capability adds cultural diagnostics that connect safety behaviour to safety metrics. Together they form a System of Intelligence built to prevent incidents, not catalogue them after the fact.

2.3 The Commercial Problem

As Joe Alner framed it during the onboarding meetings, a superior product does not sell itself in a market this conservative. Empirisys faces three compounding obstacles. First, incumbent inertia: buyers have sunk years of data and process into legacy platforms from DNV, Enablon and Sphera, and switching feels risky in a domain that is risk-averse by nature. Second, a parity illusion: as every vendor adopts the language of "AI" and "predictive analytics," a time-poor buyer struggles to tell Empirisys's domain-trained safety intelligence from a rival's marketing veneer; the differentiation is real but not self-evident. Third, a substitution threat: customers' own "DIY" AI tools built on Microsoft Copilot or ChatGPT invite a cheap, in-house "good enough" alternative that lacks the repeatability, auditability and safety-critical governance the sector requires.

THE THREE COMMERCIAL OBSTACLES

01

Incumbent inertia

Years of sunk data and process make switching feel risky in a domain that is risk-averse by nature.

02

The parity illusion

Every vendor now speaks the language of “AI”; real differentiation is not self-evident to a time-poor buyer.

03

Generic-AI substitution

Copilot and Chat GPT Enterprise invite a cheap, in-house “good enough” alternative.

Joe Alner identified three “buy triggers” where Empirisys is most likely to win: a recent fatality or major HSE failure at a prospect’s site; the arrival of a new HSE Director auditing inherited systems; or investor and board pressure for evidence-based risk management. At exactly these time-sensitive moments the sales team must prove BOOST’s superiority instantly and credibly, or the opportunity reverts to the incumbent by default.

THE THREE BUY TRIGGERS

01

A recent HSE failure

A fatality or major HSE failure at a prospect’s site puts safety systems under immediate scrutiny.

02

A new HSE Director

An incoming leader audits inherited systems and questions incumbent defaults.

03

Board pressure

Investors and boards demand evidence-based risk management.

2.4 The Project Mandate

Against this backdrop, the Statement of Work agreed with Empirisys’s leadership called for an agentic AI platform to monitor competitors and generate strategic insight on demand, with the explicit purpose of accelerating BOOST sales. Reframed as a job-to-be-done (Christensen et al., 2016), the mandate is to arm Joe Alner’s commercial function to prove, in the moment, that BOOST prevents what rivals only record, persuasively enough to convert winnable-but-stalled deals. The rest of this report sets out the primary-research diagnosis and problem tree (this section), the methodology and pivot that shaped the build (Section 3), the Porter’s and VRIO analysis of the advantage (Section 4) and the prioritised recommendations that follow (Section 5).

2.5 What BOOST Does in Operation

Before the strategic analysis, here is what the product actually does. BOOST is what Empirisys sells to customers; the Intelligence Engine in this report is the internal sales instrument built to help sell it. Product Manager Andrea Rossi and Chief Data Scientist Dr. Nyala Noe set out what BOOST does inside a customer's operation, day to day.

A high-hazard operator generates a constant stream of safety data: hazard observations, near-miss reports, permits to work, inspections and audit findings, logged by crews and HSE staff across every site and shift. Legacy Systems of Record such as Enablon, Synergi Life and Cority store this data in structured forms and produce backward-looking reports. Their weakness is twofold: they are clunky on a mobile device in the field, which suppresses both the volume and honesty of what is captured; and they report what has already happened, not what is about to.

BOOST inverts both problems. Capture is frictionless: a worker logs an observation in seconds, and BOOST integrates with capture systems such as eOBS, which lifts data quality rather than deterring reporting. More to the point, BOOST *reads* that data. As Rossi described, its AI engine interprets the free-text narrative of thousands of observations and uses topic modelling to cluster risks around the Life Saving Rules (LSRs), flagging safe and unsafe behaviours and connecting weak signals that would otherwise sit unlinked across sites. On the models Dr. Noe designs, it surfaces the *leading indicators* that precede a serious incident and produces a ranked, forward-looking picture of where the next one is most likely, early enough for an HSE Director to prevent it. That is the concrete meaning of *System of Intelligence* : not a database of what went wrong, but a layer that tells an operator where to act next.

In practice this runs on a daily cycle. As observations and near-misses arrive through eOBS, BOOST re-clusters them against the Life Saving Rules, so each morning an HSE Director opens a refreshed ranking of which sites and teams are trending toward an incident, with the underlying reports one click away. On the algorithms Dr. Noe's team maintains, a rising cluster (a run of dropped-object near-misses on one deck, say) surfaces as a leading indicator days or weeks before it would appear as an injury in a lagging report. The platform's job is to make that signal impossible to miss.

KEY TERMS USED IN THIS REPORT

Near-miss

An unplanned event that could have caused injury, damage or loss but did not: an early signal of where a real incident may follow.

System of Record vs. System of Intelligence

A System of Record stores and reports data; a System of Intelligence interprets it, prioritises it and recommends action.

Leading vs. lagging indicators

Lagging indicators count harm that has already occurred (injuries, incidents). Leading indicators (near-misses, unsafe conditions, overdue actions) point to harm *before* it happens.

COMAH

The Control of Major Accident Hazards Regulations 2015: the UK regime for sites handling dangerous substances, under which directors carry personal legal duties.

2.6 The Three Core Capabilities

BOOST delivers on this through three capabilities that work in sequence: it **predicts** where risk is concentrating, **reasons** about it inside the safety-regulatory context, and **traces** any incident to root cause on a fully auditable trail: what separates a decision-support tool from a filing cabinet.

THE BOOST CAPABILITY STACK		
01 Predictive Analytics Dashboards Rank which sites and teams carry the highest incident risk, combining lagging indicators (incidents already recorded) with leading indicators (near-misses, unsafe conditions, overdue actions), so managers act before an incident rather than after it.	02 Safety Governance AI A domain-specific model that understands HSE regulatory obligations and incident-severity frameworks. Unlike a general-purpose tool such as Microsoft Copilot, its outputs are repeatable, auditable and defensible to a regulator.	03 Incident Investigation Workflows Trace any observation or incident to its root cause along a fully auditable trail, turning an investigation from a manual reconstruction into a guided, evidenced process.

Who Uses BOOST

Frontline crew & supervisors	Capture hazard observations and near-misses in seconds on mobile. Frictionless entry is what makes the data good enough to predict from in the first place.
HSE Directors	Read the ranked risk picture, direct interventions to the highest-risk teams, and evidence prevention to the board and the regulator.
Plant & site managers	Reallocate inspections, permits and resources toward emerging risk before it becomes an incident.

How BOOST Prevents What Others Only Record

The two scenarios below are drawn from the situations BOOST is built for. Each contrasts the same events under a legacy System of Record with the outcome BOOST produces, making the abstract claim of “prevention” concrete.

01 Catching a dropped-object fatality before it happens · Offshore energy

CONTEXT

An operator runs three platforms in one field. Over six weeks, crews log a scatter of low-severity near-misses (a dropped shackle, a loose grating clip, a mishandled tool) across different shifts and platforms. Individually unremarkable, each is closed and filed by the incumbent system.

WITH BOOST

The Safety Governance AI clusters the free-text reports and detects a rising pattern of dropped-object precursors concentrated on Platform B's drilling deck. The Predictive Analytics Dashboard escalates Platform B to the top of the risk ranking on a leading-indicator basis, before any injury occurs.

OUTCOME

The HSE Director orders a targeted inspection and stand-down on that deck, finds a systemic rigging-inspection lapse, and corrects it. A potential dropped-object strike, a fatality risk, is prevented. A System of Record would have held all the same reports and flagged nothing.

02 Defending an investigation to the regulator · COMAH chemical site

CONTEXT

A COMAH-regulated site has a process-safety incident. The regulator requires the operator to evidence not just what happened, but that its safety-management system was actively managing the risk beforehand: the test that separates a defensible position from a corporate-manslaughter exposure.

WITH BOOST

The Incident Investigation Workflow reconstructs the full root-cause chain along a time-stamped, auditable trail, and the Safety Governance AI surfaces the leading indicators BOOST had flagged beforehand alongside the interventions the team logged in response.

OUTCOME

The HSE Director enters the regulatory review able to demonstrate a live, evidence-based risk-management process. This is the "liability reduction" the sales battlecards reference, shown in operation rather than asserted in a pitch.

The Prototype at the Point of Sale

Where the scenarios opposite show BOOST in a customer's operation, these two show the Intelligence Engine in the hands of Empirisys's own commercial team: the specific moments the prototype was built to win.

01 The sales battlecard pivot · Reframing a price objection

CONTEXT

An Empirisys rep is pitching BOOST to an oil & gas operator. Mid-conversation the client says they are also considering Sphera, because it is cheaper: the objection most likely to lose the deal on the wrong axis.

WITH THE HUB

The rep opens the Sphera battlecard. The governed AI's instruction is explicit: move the conversation off licence cost and onto executive liability, with the counter-points already scripted.

OUTCOME

The rep reframes: Sphera logs accidents after they happen; BOOST predicts and prevents them, protecting the board from corporate-manslaughter exposure. The deal is re-anchored on the axis where Empirisys wins.

02 The live threat scan · Sizing up a new entrant in seconds

CONTEXT

An Empirisys executive hears about a new safety-tech startup and needs to know, quickly, whether it matters, before the next leadership meeting, not after hours of manual research.

WITH THE HUB

They paste the startup's URL into the Live Threat Scan. The system scrapes the site, runs it through Anthropic's Claude, and returns a structured threat assessment in seconds.

OUTCOME

The startup is flagged as an *Emerging Threat* with a Threat Score of 75 and a summarised value proposition: decision-ready intelligence that would otherwise take hours to assemble by hand.

2.7 Project Onboarding: Diagnosing the Problem with the Empirisys Team

The problem set out above was not assumed; it was diagnosed during the project onboarding, technical briefings and client strategy sessions with the Empirisys team, supported by the internal documents they shared. Across these meetings, spanning data science, product, sales and engineering, the team explained what BOOST actually does today, why winnable deals stall, and where the causes of that commercial problem sit. Their combined account of the business is what the problem tree overleaf (Figure 1) formalises.

ONBOARDING & BRIEFING MEETINGS	
Dr. Nyala Noe Chief Data Scientist	Set out the foundational data models and how Empirisys structures its predictive risk algorithms, the basis for BOOST's forward-looking risk scores.
Andrea Rossi Product Manager	Walked through BOOST in the field: an AI engine that reads free-text safety data (observations, near-misses and audit findings), integrates with capture systems such as eOBS, and uses topic modelling to cluster risks around the Life Saving Rules (LSRs), flagging safe and unsafe behaviours. This is what makes Empirisys a predictive System of Intelligence, not a retrospective log.
Joe Alner Head of Sales & Marketing	Defined the commercial problem and the buy triggers. BOOST's buyers are senior HSE leaders who are personally and criminally liable if a site fails through negligence, and Empirisys competes against DNV, FENNEX and Enablon as well as customers' own "DIY" AI tools. The task is to prove to a liable executive that BOOST offers documented assurance they are prioritising the right risks, which a legacy filing cabinet cannot.
David Riskus & Peter Senior Full-Stack Engineer & team	Defined the MVP architecture to answer Joe's sales problem: a full-stack build (frontend, backend and PostgreSQL) whose dashboards render live strategic frameworks (Porter's, Blue Ocean and VRIO) to track competitors and evidence Empirisys's sustained advantage.

One example the team shared during these briefings captured the core of BOOST's value. An offshore client insisted that doors were not a problem on their sites. BOOST ingested their safety data and surfaced five to six finger and arm injuries, plus dozens of scattered workforce tickets on the same theme, that had fallen through the cracks of their legacy system. What legacy tools held as unconnected records, BOOST collated into a single ranked, evidenced case: the kind a leader can act on to prevent the next injury. It is this capability, sold into a UK market that Empirisys is extending into Europe and the UAE, whose durability the following analysis tests.

These briefings located the causes of the commercial problem in three places, each formalised in the problem tree overleaf. The first is structural: buyers are locked into incumbent Systems of Record by years of sunk data and COMAH-driven process, so inertia, not merit, decides many renewals. The second is perceptual: because every vendor now claims "AI," Andrea Rossi's genuinely different topic-modelling engine is invisible to a buyer who cannot see past the marketing. The third is competitive: the cheapest option is often the customer's own "DIY" tool, which looks good enough until it has to be defended to a regulator. None of these is a product weakness; each is a failure to make a real advantage legible at the moment of decision.

Figure 3 renders this diagnosis as a problem tree, tracing each root cause down to the evidence that surfaced it: the incumbent COMAH datasets locked inside Sphera and Enablon, and the offshore-doors case in which BOOST recovered five to six finger and arm injuries that legacy tools had scattered as unconnected tickets.

The Problem Tree

Reading from the roots upward: the structural root causes that produce the core commercial problem, and the consequences that follow if it is left unresolved.

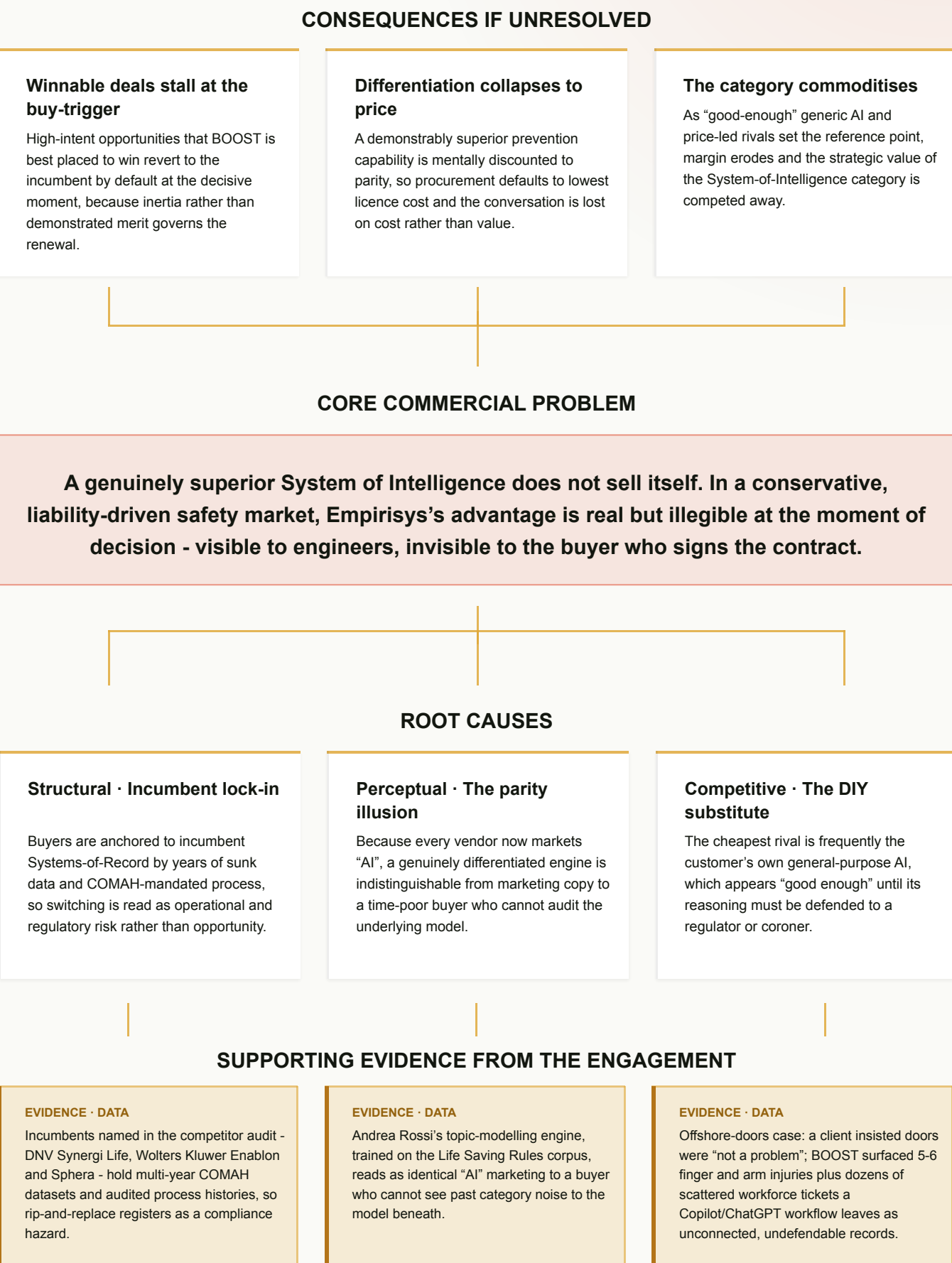


Figure 3: The problem tree, tracing the root causes of the commercial problem through to their consequences.

03

Approach

Four guiding questions, a build-to-think methodology, and the pivot that sharpened the product.

3.1 Guiding Questions for the Engagement

The engagement was structured around four guiding questions spanning the strategic, commercial and technical dimensions of the client's problem, so the analytical work and the software build would inform one another:

- 01 **Strategic durability.** Does Empirisys possess a competitive advantage over both incumbent Systems of Record and emerging general-purpose AI substitutes that is durable rather than transient, and if so, what is its source?

- 02 **Commercial enablement.** What form of on-demand competitive intelligence would most effectively equip Empirisys's sales team to convert high-intent but stalled opportunities at the precise moment a "buy trigger" occurs?

- 03 **Technical feasibility.** Can a governed, domain-specific AI system reliably translate Empirisys's strategic positioning into consistent, tactically useful counter-arguments, operationalising the company's differentiation in software, rather than leaving it to the varying skill of individual sales representatives?

- 04 **Scope and focus.** Which product capabilities actually advance the competitive-intelligence mission, and which dilute the platform's strategic clarity?

The first is answered by the Porter's and VRIO analysis in Section 4; the second and third were investigated through the prototype itself; and the fourth precipitated the strategic pivot described below.

3.2 Agile Prototyping Methodology

The project adopted an Agile, prototype-led methodology (Beck et al., 2001) rather than a linear consulting study, because the client's real problem was not fully specified at the outset and could only be surfaced through iterative dialogue and tangible artefacts. In a market as evidence-hungry as high-hazard safety, a working system is also a more persuasive proof than a written argument. Building was therefore treated as strategic inquiry: a "build-to-think" cycle of discovery, rapid construction of functional slices, live demonstration and re-scoping. The work was grounded in the onboarding meetings and technical briefings with Joe Alner, Andrea Rossi, Dr. Noe and the engineering team, and each iteration tested a hypothesis about what would actually move a deal forward.

"In a market as evidence-hungry and risk-averse as high-hazard safety, a working system is a far more persuasive proof of a claim than a written argument."

The architecture follows the design agreed with Senior Full-Stack Engineer David Riskus and Peter: a full-stack build with a frontend, backend and PostgreSQL data layer, component-driven and organised behind a “Premium Dark” intelligence-terminal aesthetic. That styling is UX-as-strategy, engineered to signal analytical gravitas to a technical, sceptical buyer. The application was built in vertical slices, complete demonstrable pages (executive dashboard, strategic frameworks, competitor query console, sales battlecards, pricing comparator and market feed) rather than horizontal layers, so each iteration produced something the client could react to.

The architecture is production-shaped even while running on seed data. The data layer runs against a live database client and falls back to mock data only when credentials are absent; the Anthropic-powered competitor endpoint streams a governed fallback when no API key is present; and the market endpoint merges live signals with seed records before degrading gracefully. The real integration points are already scaffolded and defended by schema validation and rate-limiting.

Running the demo on a curated “MVP Presentation State” (high-fidelity seed data authored to be realistic to the HSE market) was a considered trade-off, not a shortcut: it prioritised fidelity of interaction and strategy over live data while validating the concept and positioning, and decoupled the demo from the unpredictability of live scraping. The path to production is well defined: a targeted form is circulating with sales leadership to capture field inputs (the actual prices seen in RFPs and the actual reasons deals are lost to Sphera or FENNEX) which a developer substitutes into the seed arrays. This “bridge to reality” turns the prototype into a live tool through one low-friction step, and forms the backbone of the roadmap in Section 5.

A personal interest is worth noting alongside this methodology: I would only be interested in continuing to work with Empirisys beyond this engagement if it were a paid arrangement, should Empirisys be open to that. To guard against this shaping the analysis, every recommendation in this report is tied directly to the evidence set out in Sections 4 and 5, and prioritised on the Value versus Risk/Cost basis applied throughout, rather than on what would be most favourable to that possibility.

10,700

LINES OF CODE

14

APPLICATION PAGES

13

API ROUTES

37

REUSABLE COMPONENTS

3.3 The Strategic Pivot

The most consequential decision was made mid-build, in response to client feedback. Initial scope included generic CRM and lead-generation features, the modules one might expect by default from a “sales tool.” Dialogue with the client revealed the opposite: a competitor-intelligence tool becomes weaker, not stronger, when diluted with commodity functionality, because its value rests on focus and credibility. A tool that tries to be a CRM, a pipeline tracker and an intelligence platform signals none of them convincingly.

“Does this help Empirisys understand or beat a competitor?”

THE SINGLE ARCHITECTURAL TEST APPLIED TO EVERY SCREEN

The project therefore trimmed hard against that test. Capabilities that failed it (target-account radars, UK tender pipelines as first-class modules) were cut from the core, with only a lightweight market feed retained for situational awareness. This is not scope-cutting but strategy in Porter’s sense, where positioning is defined as much by the choice of what *not* to do as by what to build (Porter, 1996). The pivot preserved a sharp, ownable identity, an intelligence weapon with a single defensible purpose, whose commercial durability the next section sets out to prove.

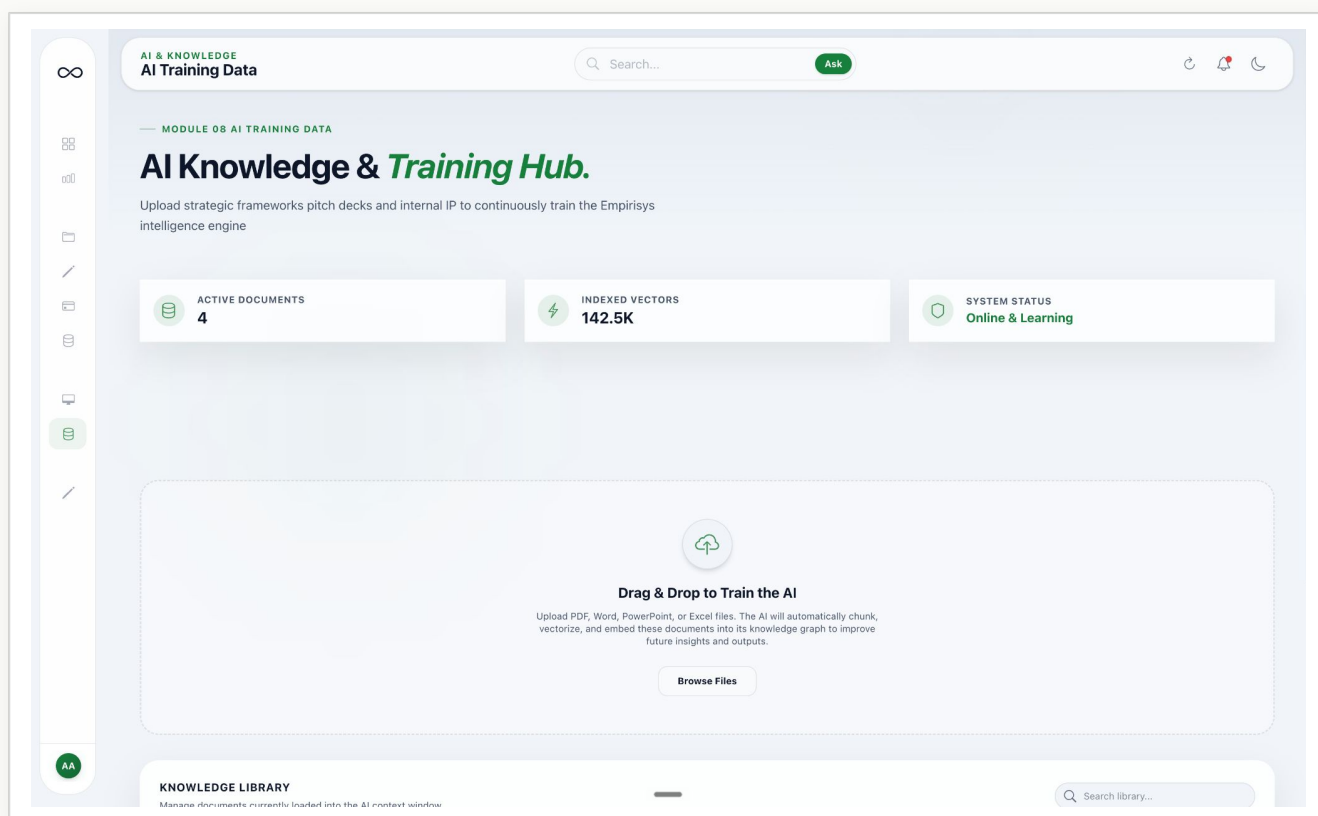


Figure 1: A working system, not a slide deck. The AI Knowledge & Training Hub ingests strategic frameworks, pitch decks and internal IP, vectorising each document (142.5K indexed vectors) into the knowledge graph that powers every downstream insight.

04

Findings & Analysis

Porter's Five Forces and VRIO reach the same verdict: the advantage is real and durable.

EXHIBIT · THE INTELLIGENCE ENGINE

Porter's & Blue Ocean, Live on Screen

The Strategic Frameworks module renders the analysis of Section 4 as live, on-screen intelligence: the five forces graded in real time, and a Blue Ocean value curve that plots exactly where Empirisys competes and legacy incumbents cannot follow.

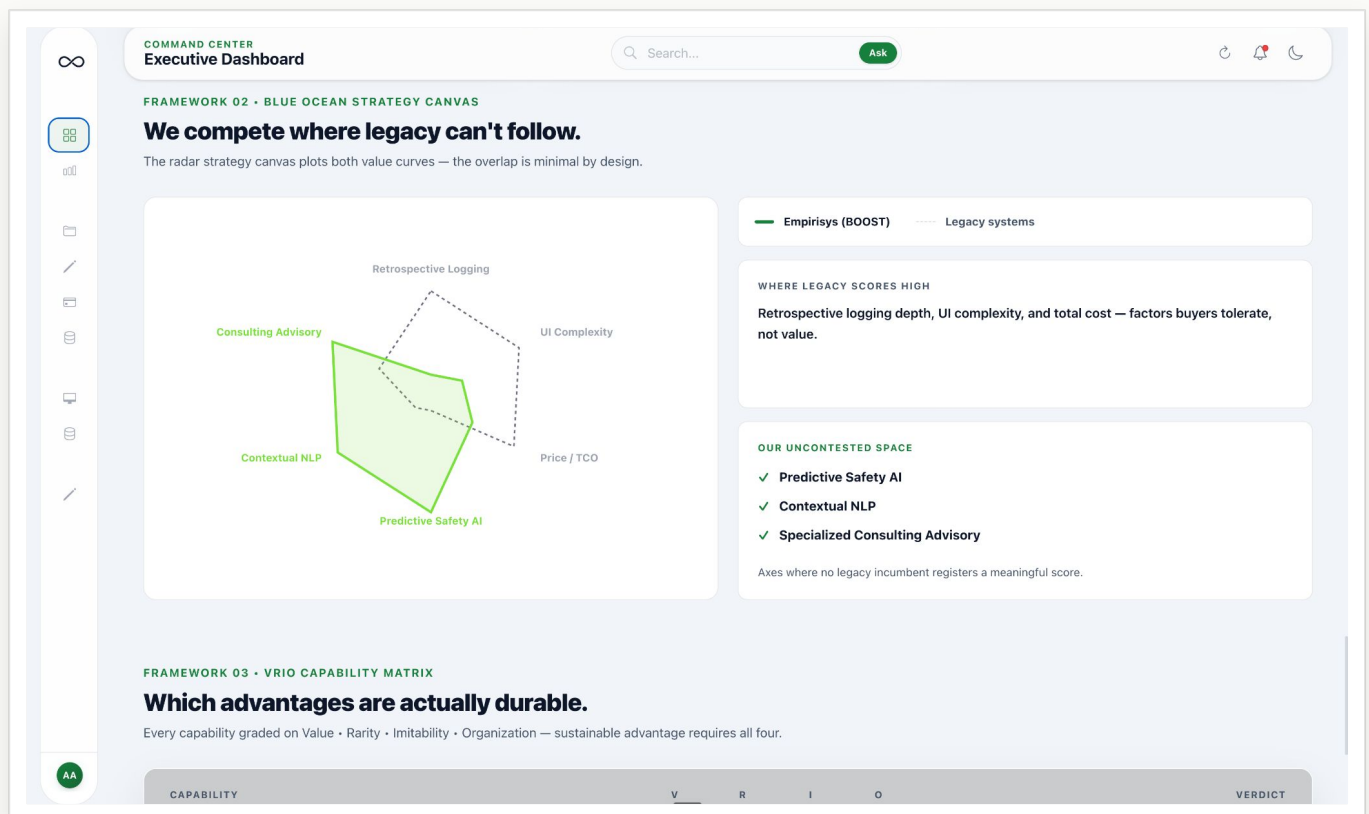
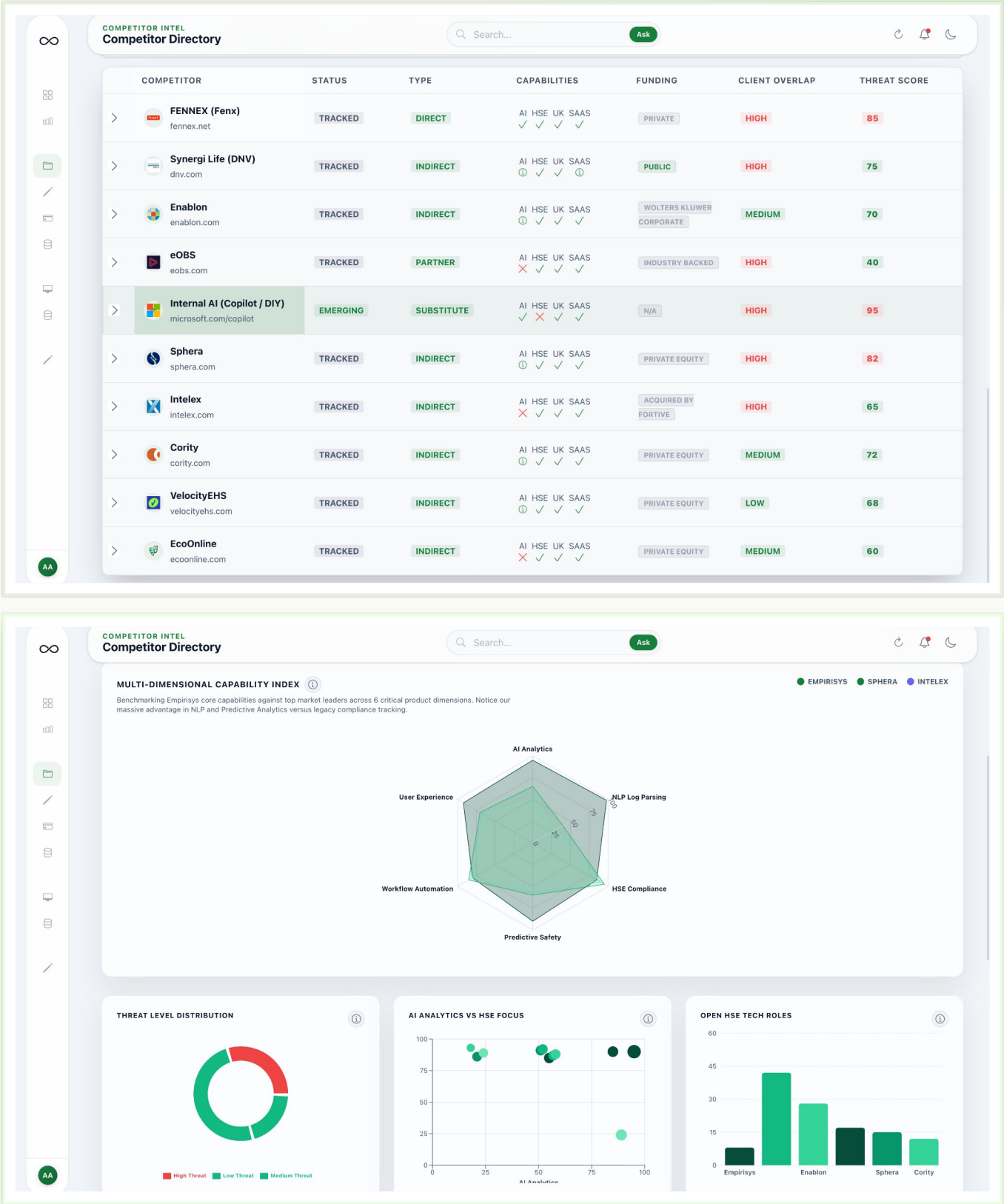


Figure 2: The Executive Dashboard, rendering Porter's Five Forces and the Blue Ocean strategy canvas from live signal extraction.

The Threat-Scoring Engine

The Competitor Directory grades every rival on a composite of AI analytics, HSE focus, UK presence and client overlap, flagging FENNEX at 85 and the generic-AI substitute at 95. The capability index benchmarks Empirisys against Sphera and Intelex across six product dimensions.



This section addresses the first guiding question: whether Empirisys's differentiation is a durable advantage rather than a head-start that incumbents or generic AI will erode. Two frameworks are applied: Porter's Five Forces assesses the structural attractiveness of the HSE software industry and which competitive posture it rewards; VRIO then tests whether Empirisys's specific resources meet the four conditions for a sustained advantage. Both also run live inside the Intelligence Engine, whose Strategic Frameworks module renders each on screen, putting the analytical case for BOOST directly in the sales team's hands.

4.1 Porter's Five Forces: An Industry That Rewards the Specialist

A surface reading of the HSE software market suggests an unattractive industry; a closer analysis shows something different (Porter, 1980): the forces are strong, but they punish the generalist and reward the focused differentiator, which is the position Empirisys has chosen.

Competitive Rivalry: High. The category is consolidated around well-capitalised incumbents (Enablon, Sphera, DNV's Synergi Life and Cority) alongside agile specialists such as FENNEX. The Hub's own threat-scoring engine, grading rivals on AI analytics, HSE focus, UK presence and client overlap, ranks the largest incumbents most dangerous while flagging FENNEX at 85, owing to near-identical messaging and shared territory in the Aberdeen offshore sector. Yet this rivalry is fought almost entirely on one axis, retrospective data capture with a retrofitted AI layer, leaving the adjacent axis of predictive, preventive intelligence comparatively uncontested. The platform's Blue Ocean canvas (Kim and Mauborgne, 2005) makes this visible: where incumbents cluster (logging depth, interface complexity, cost), Empirisys scores low; where Empirisys scores high (predictive safety AI, contextual NLP, advisory), no legacy incumbent registers.

85

FENNEX THREAT SCORE

Threat of New Entrants: Moderate. Open-source language models and commodity cloud have lowered the barrier to building "a safety AI." But two barriers hold: domain credibility, since a market where software errors contribute to fatalities is inhospitable to unproven tools; and the entanglement of enterprise HSE contracts, where accumulated data and COMAH compliance create switching costs that shield incumbents. Entry at demonstration level is easy; entry at enterprise-grade, safety-critical credibility is hard. The net threat is moderate.

Threat of Substitutes: High and Rising. The most dangerous force is external: the customers' own "DIY" AI on Microsoft Copilot or ChatGPT, a cheaper in-house "good enough" alternative that threatens to commoditise HSE analytics. This is the single greatest risk Empirisys faces, which is why the Hub's competitor set explicitly includes an "Internal AI (Copilot / DIY)" entry. Empirisys's counter is encoded in the governed AI: generic tools lack the repeatability, auditability and safety-critical governance high-hazard sites demand, where an unaccountable answer is not just unhelpful but dangerous. The counter is defensible, but the threat is structural, and it shapes the recommendations that follow.

Bargaining Power of Buyers: Moderate to High, with a decisive qualification. Buyers are sophisticated and concentrated, and procurement runs through formal, price-transparent tenders, all of which elevate buyer power. The qualification is the crux of the whole thesis: the buyer's personal and criminal liability sharply reduces price sensitivity for any solution that credibly cuts their exposure. Someone weighing a marginal licence fee against a corporate-manslaughter investigation does not behave as a price-driven commodity buyer. Buyer power is thus high on process but attenuated exactly where Empirisys is strongest (demonstrable prevention), and the Hub's pricing and battlecard modules are built to exploit that gap, reframing the conversation from unit cost to total cost of ownership and liability.

Bargaining Power of Suppliers: Low. Empirisys's principal "suppliers" are the foundation-model and cloud providers. Because that layer is commoditised and multi-sourced, dependency is low: the prototype integrates two independent inference providers to insulate itself from any one vendor's pricing or availability. The scarce input is not the model but the proprietary safety data and domain expertise, which Empirisys owns. Supplier power over the strategic core is minimal.

Synthesis. Read together, the five forces back Empirisys's strategic choice. In an industry with strong forces, sustainable returns accrue to firms that differentiate or focus, not those that chase cost leadership. Intense rivalry and rising substitution both attack the commoditised, retrospective axis of the market, while the liability dynamic and regulatory complexity carve out a protected niche for a vendor credibly positioned on prevention. The forces punish the "me-too" System of Record; they reward the focused System of Intelligence.

"The forces punish the 'me-too' System of Record; they reward the focused System of Intelligence."

4.2 VRIO: Locating the Source of Sustained Advantage

Porter's establishes that the industry rewards differentiation; VRIO tests whether Empirisys holds the resources to sustain it. Barney's framework (Barney, 1991; Barney and Hesterly, 2015) checks each resource against four conditions in sequence (Value, Rarity, Imitability and Organisation), with a sustained advantage arising only where all four hold. Applied to Empirisys's core capabilities, it yields the assessment below, which mirrors the verdicts in the platform's own VRIO module.

CAPABILITY	V	R	I	O	VERDICT
Domain-trained contextual NLP engine (BOOST)	✓	✓	✓	✓	Sustained advantage
Embedded consulting advisory layer	✓	✓	✓	✓	Sustained advantage
Real-time signal pipeline	✓	✓	✗	✓	Temporary advantage
Competitors' bolt-on AI / legacy vendor APIs	✓	✗	✗	✗	Competitive parity

Figure 5: VRIO capability assessment (V: Valuable, R: Rare, I: Inimitable, O: Organised).

BOOST's domain-trained language model, the topic-modelling engine Andrea Rossi described, is *valuable* because it turns unstructured safety narratives into prioritised, preventive action, neutralising the incumbents' "we capture but do not interpret" weakness. It is *rare* because no rival holds the combination: incumbents own retrospective databases but not safety-tuned models, while general-purpose AI owns models but not safety-domain tuning. It is *costly to imitate* because it rests on years of proprietary safety data and consulting engagements, protected by causal ambiguity: a competitor cannot reverse-engineer why the interpretation is trustworthy, nor conjure safety-critical repeatability by pointing a generic model at the problem. Finally, Empirisys is *organised* to exploit it, through a model-and-database architecture built for contextual understanding rather than the rigid schemas of the incumbents. All four conditions are met: this is a sustained competitive advantage.

The embedded advisory layer passes the same test differently. Its inimitability comes from social complexity and reputation (Barney's classic sources of durable advantage), built over years of safety-consulting relationships that cannot be bought or replicated at speed. Where rivals sell either software or consulting, Empirisys fuses expert judgement into the product; the Intelligence Engine is itself an expression of this, its governed prompts encoding the firm's advisory doctrine so expertise scales beyond the individual consultant.

The real-time signal pipeline, though, grades as only a *temporary* advantage. It is valuable and, for now, rare, but not inimitable: the scrapers and feeds behind it are replicable engineering, not a moat. That distinction is the most commercially useful finding here, because it tells Empirisys exactly where not to over-invest, a conclusion carried straight into the recommendations.

The incumbents' bolt-on AI, meanwhile, clears only the first hurdle. It has some value, but it is not rare (every vendor offers it), not inimitable (attaching an AI layer to a database is easily copied) and not really exploited by organisations still built as Systems of Record. It delivers, at best, competitive parity.

4.3 Convergent Conclusion

The two frameworks point to the same conclusion. Porter's shows the HSE software industry structurally rewards a focused differentiator and penalises the commoditised generalist. VRIO shows Empirisys holds exactly the inimitable resources (a domain-trained language model and a socially complex advisory heritage) needed to occupy and defend that position over time. The advantage rests on the two capabilities hardest to copy (accumulated domain data and reputational expertise), not on the replicable signal pipeline.

“Empirisys possesses a sustained competitive advantage; its source is the fusion of safety-domain intelligence and embedded expertise.”

The answer to the first guiding question is therefore affirmative: Empirisys holds a sustained competitive advantage, sourced in the fusion of safety-domain intelligence and embedded expertise, and the Intelligence Engine is the instrument that makes that advantage visible, repeatable and saleable at the point of decision.

Figure 6 mirrors these findings as a solution tree, mapping the three VRIO-verified foundations - the domain-trained NLP engine, the embedded advisory heritage and the proprietary data moat - onto the commercial outcomes the Intelligence Engine is built to deliver at the point of sale.

The Solution Tree

The mirror of Figure 3: the inimitable foundations of Empirisys’s advantage support the core solution, which in turn delivers the commercial outcomes the Intelligence Engine is built to produce.

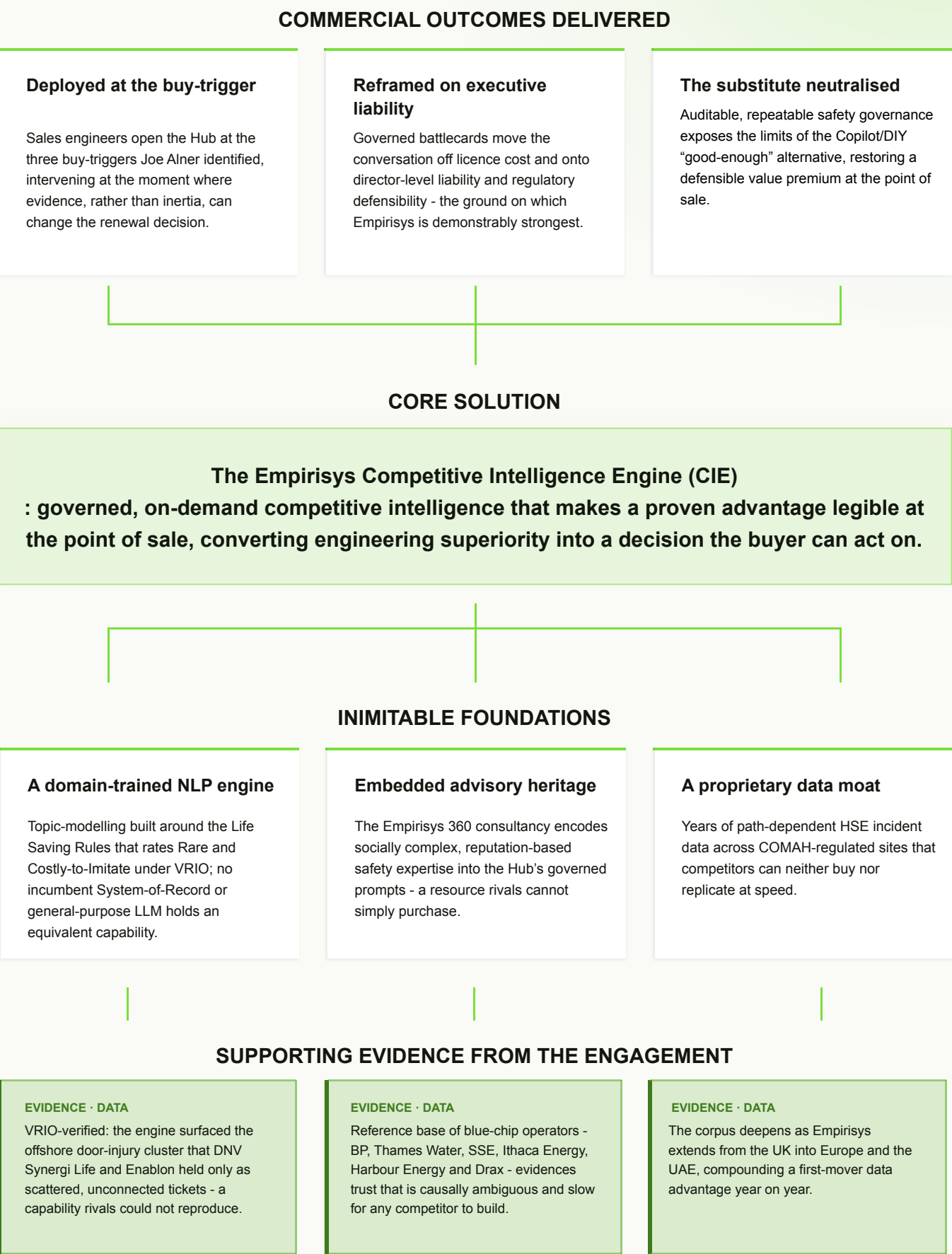


Figure 6: The solution tree, mapping the foundations of Empirisys’s advantage to the outcomes the Intelligence Engine delivers.

05

Recommendations

Put the proven advantage to work now, fund the moat with intent, and decline the replicable.

Section 4 points to one priority: Empirisys should concentrate investment on the capabilities that are truly inimitable (its domain-trained model and embedded advisory heritage), resist over-building the replicable, and urgently insulate itself against the generic-AI substitute. The recommendations below are mapped onto a Value versus Risk/Cost matrix: the vertical axis measures each initiative’s *value* to accelerating BOOST sales, the horizontal axis its *risk and cost* to deliver. The four quadrants convert strategy into sequence.

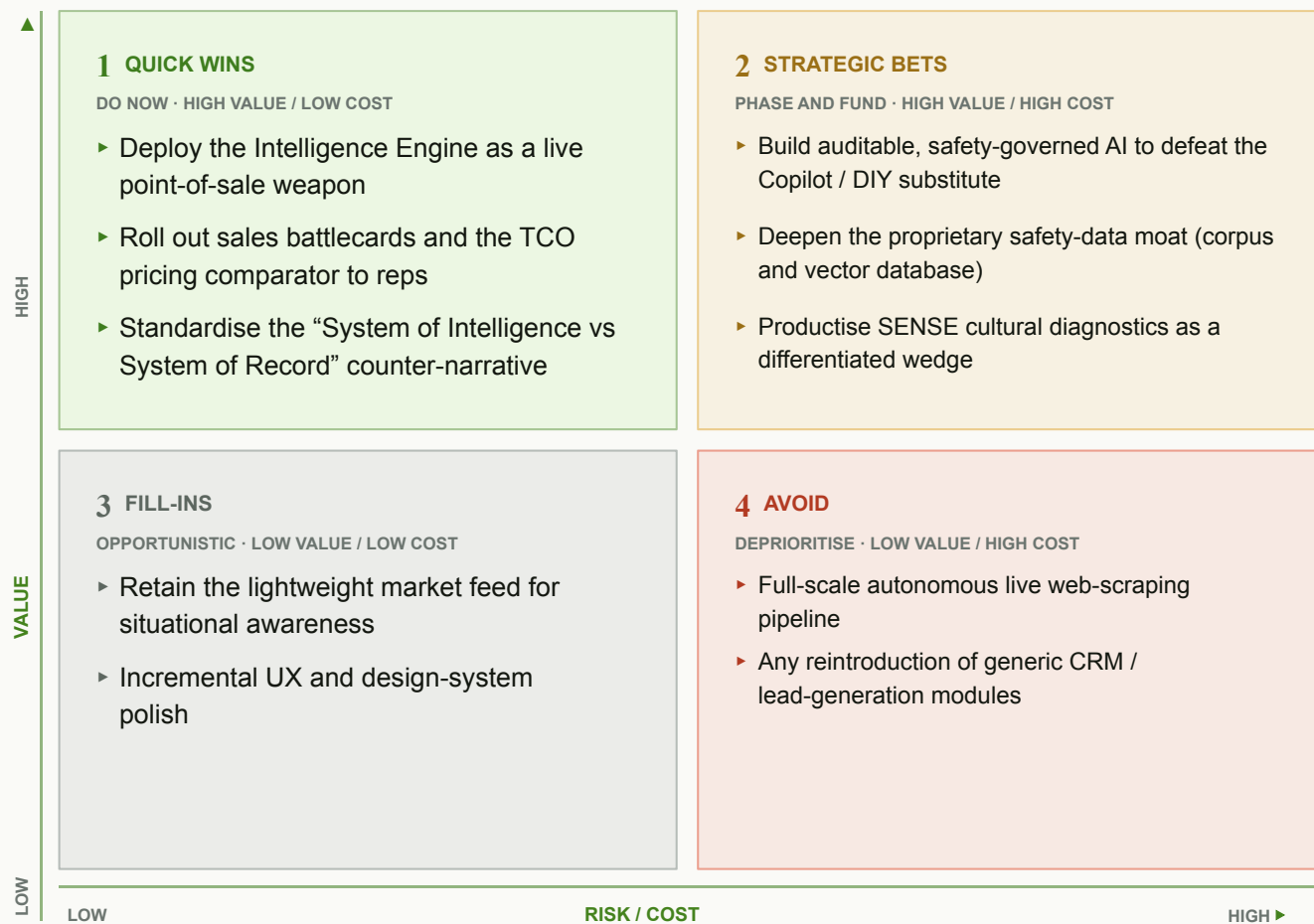


Figure 7: Value versus Risk/Cost prioritisation matrix.

5.1 Quadrant 1: Quick Wins (High Value, Low Cost)

The most important recommendation is to **deploy the agentic Intelligence Hub to Joe Alner’s sales team now** , putting the competitive advantage the analysis proves to work. The value is immediate and the assets exist: the Next.js application, its governed AI console, its competitor tracking (on HSE focus, UK presence and client overlap), its Live Threat Scan and its natural-language intelligence assistant are all built. Going live requires only the “bridge to reality” already in motion: a form with sales leadership capturing the real RFP prices and the real reasons deals are lost to Sphera or FENNEX, which a developer drops into the seed arrays. In the field, reps use the Hub’s live dashboards and battlecards at the exact buy-trigger moment to reframe the pitch off software cost and onto *executive liability* , letting the AI prove that BOOST prevents what legacy systems only record. The Total Cost of Ownership comparator reinforces this, exploiting the Five Forces finding that liability attenuates price sensitivity; and the counter-narrative should be standardised across sales and marketing, formalising the positioning already encoded in the platform’s server-side prompts.

5.2 Quadrant 2: Strategic Bets (High Value, High Cost)

These sustain the advantage over a multi-year horizon and warrant planned funding and phasing. The most urgent is to neutralise the substitution threat structurally, investing in the auditable, deterministic safety-governance features Section 4 identified as generic AI's blind spot: audit trails, repeatable outputs and demonstrable controls a Copilot deployment cannot replicate. In parallel, Empirisys should deepen its proprietary data moat, expanding the safety-domain training corpus that underpins the inimitable VRIO capability: capital-intensive, but the one resource competitors cannot buy or copy. Extending SENSE as a productised cultural-diagnostic wedge adds a further differentiated entry point no System of Record can match.

5.3 Quadrant 3: Fill-ins (Low Value, Low Cost)

These are worth retaining opportunistically but should never absorb disproportionate attention. The lightweight market feed provides useful situational awareness, and incremental UX polish sustains the credibility-signalling "intelligence terminal" aesthetic. Neither is a source of advantage, and both should yield to any Quadrant 1 or 2 work competing for the same resources.

5.4 Quadrant 4: Avoid (Low Value, High Cost)

This quadrant is where the analysis is most decisive, and it backs the pivot. Building the fully autonomous live web-scraping pipeline to production scale would be costly, yet VRIO graded exactly this capability as only a *temporary* advantage: replicable engineering, not a moat. That the discovery endpoint remains a placeholder is the right call, not an oversight, and it should stay deprioritised in favour of moat-building. Equally, any reintroduction of generic CRM or lead-generation modules must be resisted: the pivot removed them because they dilute clarity for no defensible return, and choosing what *not* to do is itself a source of advantage.

5.5 Strategic Sequencing

The matrix sets the order of play: act now on Quadrant 1, converting the prototype into live commercial impact at minimal cost; fund and phase Quadrant 2 to defend against the substitute and deepen the data moat; maintain Quadrant 3 lightly; and decline Quadrant 4. This directs scarce resources to the inimitable rather than the replicable (the central lesson of VRIO), and aims the firm's commercial energy at the moment and the message where its advantage is most persuasive.

5.6 Implementation: Deploying the Agentic Hub to the Sales Floor

Because the headline recommendation rests on the prototype, this section states plainly what has been built and how Joe Alner's team would use it, so the recommendation is understood as an operational plan rather than an aspiration. The Empirisys Competitive Intelligence Engine (CIE) is a full-stack web application that acts as an AI agent for competitive analysis and lead scoring. It tracks every named rival (DNV, FENNEX, Enablon) and the "DIY" internal-AI substitute across the dimensions Empirisys actually competes on: depth of safety-AI analytics, HSE focus, UK presence and overlap with Empirisys's target clients. Its two working front doors for a rep are a *Live Threat Scan*, which assesses a competitor from its own web presence on demand, and an embedded intelligence assistant that answers natural-language questions about the competitive field in seconds.

Deployment is a sequence, not a switch. First, sales leadership complete the "bridge to reality" data step, replacing seed values with the real prices seen in live RFPs and the real reasons deals are lost, so the Hub speaks in the numbers a buyer will recognise. Second, reps are trained to open the Hub at the three buy-trigger moments the diagnosis identified (a recent HSE failure, a newly appointed HSE Director, or board pressure for evidence-based risk management), because these are the windows where a decisive intervention changes the outcome. Third, each rep works from the battlecard and pricing views that match the specific rival in the room.

In practice this changes the conversation on the axis where Empirisys wins. When a prospect raises Sphera's lower licence cost, the rep opens the Sphera battlecard, and the governed AI moves the discussion off price and onto executive liability with the counter-points already scripted: Sphera records accidents after they happen; BOOST predicts and prevents them, protecting the board from a corporate-manslaughter exposure. When a new competitor surfaces, the Live Threat Scan returns a structured assessment before the next leadership meeting rather than after hours of manual research. And when a newly appointed HSE Director is auditing inherited systems, the rep can ask the assistant, in plain language, how BOOST compares with DNV for a COMAH operator, and read back an evidenced, on-doctrine answer on the spot. In each case the AI is proving, in the moment, that BOOST prevents what a legacy System of Record only records.

Two disciplines keep the deployment honest. The system prompts are governed centrally, so every rep delivers the same on-doctrine positioning rather than improvising it, and success is measured narrowly: win-rate on deals that reach a buy-trigger, and the proportion of those deals reframed onto liability rather than cost. The underlying technical detail (the application architecture agreed with David Riskus and Peter, the PostgreSQL data layer, and the server-side model orchestration) is documented in Appendix A, so that the body of this report can stay focused on the commercial decision the Hub is built to serve.

06

Conclusion

From abstract differentiator to demonstrable proof: an ownable, defensible, saleable position.

This report addressed a specific problem, diagnosed during the project onboarding and technical briefings with the Empirisys team: BOOST is a strongly differentiated safety-intelligence platform, yet operates in a conservative, liability-driven market where a superior product does not sell itself. Incumbent inertia, a pervasive “parity illusion” and a rising substitution threat from customers’ own AI all obscure or erode its real advantage. The mandate, as Joe Alner framed it, was to arm the commercial function to prove, at the decisive moment, that BOOST prevents what rivals only record.

The response was not a document but a working instrument. Built through an Agile, prototype-led methodology on the architecture agreed with David Riskus and Peter, The Empirisys Competitive Intelligence Engine (CIE) is a production-shaped full-stack application whose governed, multi-model AI encodes the firm’s doctrine directly into software, so differentiation is delivered by the system rather than the variable skill of individual reps. A mid-build pivot, in response to client feedback, sharpened it against one test, “does this help Empirisys understand or beat a competitor?”, trading breadth for a defensible identity.

The analysis confirmed this identity is commercially durable. Porter’s showed the industry structurally rewards the focused differentiator and punishes the generalist, while buyers’ personal and criminal liability attenuates price sensitivity exactly where Empirisys is strongest. VRIO located the advantage in the two resources hardest to copy (a domain-trained language model and a socially complex advisory heritage), both meeting all four conditions for a *sustained* advantage, against competitors’ bolt-on AI, which registers only as parity. The replicable signal pipeline was candidly graded temporary, a finding that shaped where the firm should, and should not, invest.

Those priorities were crystallised in a Value versus Risk/Cost matrix. Empirisys should act now on quick wins: deploying the Hub to the sales team via the “bridge to reality” step and using its battlecards and TCO comparator to reframe deals around liability, not licence cost. It should fund and phase the strategic bets that defend the advantage (auditable, safety-governed AI to defeat the generic-AI substitute, and continued deepening of the data moat), while declining the capital-intensive, low-moat work the pivot already rejected.

The Intelligence Engine, then, achieves more than a product demonstration. It de-risks the commercial strategy, converts an abstract differentiator into demonstrable proof, and equips Empirisys leadership with a competitive position that is not only real but defensible, and, through this prototype, finally saleable at the point of decision.

Declaration on the Use of Artificial Intelligence

In line with the University's policy on the acceptable use of generative artificial intelligence, this declaration sets out, transparently and in full, where and how AI tools were used in the production of this report and the accompanying software prototype. The intellectual substance of this work (the analysis, the judgement and the conclusions) is my own.

1 ORIGINAL INTELLECTUAL WORK

The analytical content of this report is entirely my own. This includes my analysis of the project onboarding meetings, technical briefings and client strategy sessions with the Empirisys team (Dr. Nyala Noe, Andrea Rossi, Joe Alner, David Riskus and Peter), together with the internal documents they provided; the diagnosis of the commercial problem and its causes; the design of the problem and solution trees; the application of the strategic frameworks (Porter's Five Forces, VRIO and Blue Ocean); and the prioritised commercial recommendations. The argument, its structure and every conclusion drawn are my original work.

2 AI AS A SOFTWARE DEVELOPMENT AID

In building the software prototype (The Empirisys Competitive Intelligence Engine), I used AI programming assistants (Codex and Antigravity) strictly as a development aid, in the same role a developer uses code completion or documentation. Their use was limited to generating boilerplate code and scaffolding standard React components. All architectural decisions, the data model, the API design, the scraping and retrieval logic, and the integration of these parts into a working system were specified, directed and verified by me. The design of the system is mine; the assistants accelerated its typing, not its thinking.

3 AI AS A PROOFREADING TOOL

A large language model was used solely as a copy-editing and proofreading tool, to refine grammar, sentence structure and formatting in the final manuscript. It was not used to generate ideas, arguments, analysis or source material. Every sentence expresses my own reasoning; the tool assisted only with the clarity and correctness of its expression, and all output was reviewed and approved by me.

I confirm that this use of AI is consistent with the University's academic-integrity regulations, that it has been declared honestly and in full, and that I take complete responsibility for the entire content of this submission.

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Appendix A (Statement of Work)

Competitor Analysis

A research project mapping the competitive landscape of the UK process safety and HSE analytics sector.

Project title	Competitor analysis: researching, mapping and delivering a structured competitive intelligence framework for Empirisys Ltd in the UK process safety and HSE analytics sector.
Student	Aakash Arora
Programme	MBA with Artificial Intelligence, Cardiff Business School, 2025 to 2026
Client	Empirisys Ltd
Client contacts	Joe and Dawn
Co-founders	Gus Carroll and Peter
Academic supervisor	Professor Omer Rana
Duration	Five weeks
Date	June 2026

01 Background and Client Brief

Empirisys Ltd is a UK process safety and HSE (Health, Safety, Environment) analytics company. Its core products are BOOST, an AI analytics engine for unstructured HSE data, SENSE, a cultural and leadership diagnostics platform, and Empirisys 360, a consultancy framework. Clients include BP, Thames Water, SSE, Ithaca Energy, Harbour Energy and Drax. I was selected for the competitor analysis workstream through the Cardiff Business School MBA Accelerator Programme. The client brief reads:

“The project will be centred around AI-assisted competitor analysis and market scanning, using Agentic AI to firstly build out a view of the current state of our competitive landscape, and then to alert us around new activity, potential threats and opportunities. Key deliverables would ideally be a pilot or prototype of the service that we could then put into production.”

This document defines my scope, methodology and deliverables. My project focuses exclusively on competitor analysis and remains independent from the parallel market research workstream, which is assigned separately to another student under a different brief.

02 Research Aim and Objectives

Primary research question: how can a structured research methodology be applied to build a scalable and continuously updated competitor intelligence framework for Empirisys Ltd in the UK process safety and HSE analytics sector?

#	Objective
1	Map the competitive landscape, identifying all direct and indirect competitors.
2	Develop structured profiles for a minimum of 10 key competitors across positioning, capabilities, focus and threat level.
3	Identify competitive threats, market gaps and strategic opportunities for Empirisys.
4	Design and apply a structured research method to gather, process and synthesise competitor intelligence.
5	Produce a reusable competitor intelligence framework, and a working pilot that demonstrates how it can be queried and maintained after the project concludes.

03 Research Methodology

This project is research-led. The primary deliverable is a structured competitor intelligence framework grounded in original research. Established analytical tools are used to assist with data gathering, pattern recognition and synthesis, in the same way a researcher might use analytical software to process data.

Phase	What it involves
1 . Identification	Mapping all players from public sources including company websites, LinkedIn, Crunchbase, Companies House, G2 and industry publications, separating direct from indirect competitors.

Phase	What it involves
2 . Profiling	Documenting each competitor across overview, products, market positioning, threat assessment, and content and marketing activity.
3 . Analysis	Synthesising the data into comparative positioning analysis, market gap identification and an assessment of Empirisys relative to key competitors, with every output verified against the original research.
4 . Framework and pilot	Structuring findings into a reusable framework and a working demonstration that answers competitor queries, as a proof of concept for ongoing monitoring.
5 . Reporting	Compiling findings, framework and pilot into a structured deck with guidance on how Empirisys can use and maintain it.

04 Scope

Within scope	Outside scope
Competitor landscape mapping, minimum 10 companies	Market research and industry trend analysis, a separate workstream
Structured competitor profiling across all dimensions	Marketing strategy recommendations, a separate workstream
Competitive threat and opportunity analysis	Internal product development, pricing or contract decisions
Research methodology, designed and applied by me	Legal or regulatory compliance advice
Competitor intelligence framework and pilot demonstration	Post-project maintenance, and any work from or dependent on the parallel workstream

05 Deliverables

#	Deliverable	Phase
1	Competitor landscape map of all direct and indirect competitors	Identification
2	Structured profiles for a minimum of 10 competitors	Profiling
3	Competitive threat and opportunity analysis for Empirisys	Analysis
4	Competitor intelligence framework with documented methodology	Framework
5	Working demonstration of the competitor intelligence framework	Framework
6	Structured research deck and handover documentation	Reporting

06 Research Timeline

Week	Phase	Focus
Week 1	Competitor identification	Map direct and indirect competitors, confirm shortlist with the client
Week 2	Competitor profiling	Research and document all 10 competitor profiles
Week 3	Competitive analysis	Synthesise findings, produce threat and opportunity analysis, verify against source data
Week 4	Framework and pilot	Design the framework, build and test the pilot
Week 5	Deck and handover	Compile the deck, prepare handover, present findings

07 Independence and Academic Integrity

This is a standalone academic deliverable. While Empirisys has assigned a parallel market research workstream to another MBA student, my competitor analysis workstream is entirely separate. The two address different research questions, draw on different data sources and produce different outputs.

I will not use, reference or build upon any research, data or output produced by the parallel workstream. All research, analysis and framework design here is my own original work, and any AI tools are applied as instruments under my own direction. This document has been written independently and reflects my own scope of work.

08 Stakeholders

Stakeholder	Role	Involvement
Aakash Arora	Student researcher	Sole researcher and author of all deliverables
Gus Carroll	Co-founder, Empirisys	Strategic direction and final sign-off
Peter	Co-founder, Empirisys	Product and positioning context
Joe	Client contact	Primary contact for the brief and research validation

Stakeholder	Role	Involvement
Dawn	Client contact	Market context and research priorities
Professor Omer Rana	Academic supervisor	Oversight, methodology guidance and sign-off

09 Risks and Success Criteria

Risk	Impact	Mitigation
Public competitor data is incomplete	Medium	Triangulate across multiple sources, flag gaps transparently
Client availability for validation is limited	Low	Asynchronous validation by email, sessions kept to 30 minutes
AI analysis produces unsupported claims	Medium	All outputs verified against original research before inclusion
Scope expands beyond competitor analysis	High	Strict scope boundary maintained, market research excluded

Success criteria: Professor Rana confirms a clear scope of work; the client confirms the competitor map and profiles are accurate and useful; the framework is structured and reusable; the demonstration answers competitor queries using the research data; and the final deck is delivered on time as an independent piece of academic work.

10 Immediate Next Actions

#	Action	Timing
1	Getting the approval on revised SOW by Professor Umer Rana	Before 20th June
2	Confirm the competitor shortlist with Joe and Dawn	Week 2
3	Begin competitor identification and landscape mapping	Week 2
4	Confirm two 30 minute weekly check-in sessions with the client	Every week, Monday and Wednesday
5	Confirm research tool access with Empirisys	Week 1 & 2
6	Academic report writing	Week 2 onwards

Signatures

Client representative		Academic supervisor	
Name	Peter Sueref	Name	
Date	22/06/2026	Date	
		Signature	
Signature			

Appendix B, Ethics Compliance

CARBS – BST906 Accelerator Project (MBA)

Dissertation Ethics Form

1. PROJECT SUMMARY				
Project Title:		The Empirisys Competitive Intelligence Engine (CIE) A Sustained Competitive Advantage in the HSE Software Market		
a) Please provide a summary of your project, including its aims and the methods you will be using:				
This consultancy engagement, undertaken for Empirisys Ltd (a safety-intelligence software vendor to high-hazard industries) under an agreed Statement of Work, aimed to test whether Empirisys's product differentiation is commercially durable and to convert that differentiation into a sales-ready asset. Using Porter's Five Forces, VRIO and Blue Ocean strategy frameworks, the project analysed Empirisys's competitive position against incumbents (DNV, Sphera, Enablon, Cority) and built a working prototype, the Empirisys Competitive Intelligence Hub, using an Agile, prototype-led methodology. Evidence was gathered through project onboarding and briefing meetings with Empirisys staff, internal company documents shared under the engagement, and secondary desk research on competitors and the UK HSE software market.				
b) If working with human subjects, please confirm your participant group and how you will identify and recruit such participants:				
Not applicable.				
c) How many participants are you aiming to recruit?	10 or less ☐	11-20 ☐	21-49 ☐	50+ ☐
d) Please describe the consent procedures/arrangements for the project e.g. how will informed consent be obtained from participants (where relevant)?				
Not applicable.				
e) What do you consider to be the ethical issues or considerations specific to your project? How do you intend to address these?				
The principal consideration was confidentiality of Empirisys's commercial, product and strategic information. This is addressed by using it solely for the agreed engagement and academic assessment, attributing views to stakeholders' job titles with the client's knowledge, and sharing no information beyond what Empirisys agreed to disclose under the Statement of Work. No personal, sensitive or special category data was collected; the primary risk is commercial rather than to individual wellbeing, and it is managed through the scope agreed in the Statement of Work and supervisor oversight.				
f)	Do you have any potential conflicts of interest to declare? <i>Information and guidance on conflicts of interest is contained in the Research Integrity Online Training Programme and the Research Integrity and Governance Code of Practice.</i>	Yes	No	
		✓	□	
If you have answered 'yes' to f, please confirm the action you propose to take to address such conflicts:				
I would only be interested in continuing to work with Empirisys beyond this engagement if it were a paid arrangement, should Empirisys be open to that. To address this, every recommendation in this report is tied directly to the evidence gathered and the Value versus Risk/Cost basis applied throughout, and this interest has not influenced the report's content, conclusions or recommendations.				

2. IS YOUR PROJECT ELIGIBLE FOR SUPERVISOR-ONLY REVIEW?		
Does your project involve any of the following:	Yes	No
a) Activity within, or involving, the NHS or health and social care systems (e.g. patients, service users, staff, resources or data).	☐	☒
b) The collection or use of Human Tissue , as defined by the Human Tissue Act 2004	☐	☒
c) The use or involvement of animals.	☐	☒
d) The subject areas of terrorism, extremism or radicalisation. <i>Note: these terms are defined in the University's Security Sensitive Research Policy.</i>	☐	☒
e) [Primary data collection without the prior consent of participants* <i>*The following are specifically excluded from this criteria: (1) observation of people in a public space where no identifiable information is being captured; (2) use of publicly available information or secondary data where no identifiable information is being captured.]</i>	☐	☒
f) [An element of deception, including covert activity]	☐	☒
g) [Children under the age of 18 or 'at risk' (vulnerable) adults or groups]	☐	☒
h) [Topic areas likely to be considered highly sensitive for participants. <i>Note: examples include sexual behaviour, illegal activities, political, religious or spiritual beliefs, race or ethnicity, experience of violence, abuse or exploitation, and mental health.]</i>	☐	☒
i) [Access to records of a sensitive or confidential nature, including ' Special Category Data ' or criminal offence data .]	☐	☒
j) [Permission of a gatekeeper for initial or continued access to participants. <i>Note: this includes participants in custody and care settings, or in communities where access is not possible without the permission of another adult (such as a family member or community leader).]</i>	☐	☒
k) [Intrusive or invasive procedures <i>Note: examples include the administration of substances, vigorous exercise, procedures which may involve pain or more than mild discomfort to participants (including the risk of psychological distress or anxiety – more than the risks encountered in everyday life).]</i>	☐	☒
l) [More than a minimum risk of harm to the safety and wellbeing of participants and/or the student conducting the project]	☐	☒
m) Any other activity which clearly falls outside of the parameters of the Module Ethics Framework or raises other material ethical concerns?	☐	☒
If you have selected “Yes” to any of the above, your project is NOT eligible for Supervisor sign-off and must be reviewed by the SREC. Please <u>do not</u> continue with this form. You must instead complete the SREC’s ‘Application Form for Ethical Review’. If you have selected “No” to all of the above, your project is eligible for Supervisor sign-off and you must complete the rest of this form. Your project <u>does not</u> require a SREC application.		

3. DECLARATIONS – For use where project <u>IS</u> eligible for Supervisor sign-off.	
STUDENT DECLARATIONS	
a. The information in this form is accurate to the best of my belief. I take full responsibility for it.	✓
b. I have read, and will adhere to, the Module Ethics Framework .	✓
c. For research using secondary data and publicly available data, I have read and will adhere to the Framework for the ethical review of research using Secondary Data or Publicly Available information .	✓
d. I have completed the University's Research Integrity Training and believe I have the necessary skills and knowledge to conduct the project as proposed.	✓
e. I am familiar with the Ethics Policy for Human Research , Research Integrity and Governance Code of Practice and Academic Integrity Policy and will abide by all standards relevant to my project.	✓
f. I [have submitted] [will submit] all final version participant-facing documents and data collection tools to my Supervisor for approval prior to commencing data collection.	✓
g. I will notify my Supervisor of any amendments to my project and seek approval of any substantial changes to participant-facing documents or data collection tools.	✓
h. I have read the risk assessment (Annex 4) and have considered it for my research.	✓
i. I will notify my Supervisor of any amendments to my project and seek approval of any substantial changes to participant-facing documents or data collection tools.	✓
j. I will notify my Supervisor of any ethical issues that arise during the project.	✓
Signed: 	
Print name: Aakash Arora	
Date: 21st July 2026	

SUPERVISOR DECLARATIONS	
a. I have reviewed this form and I am satisfied that the project, as proposed, meets ethical standards.	☐
b. I [have reviewed] [will review] all participant-facing documents and data collection tools prepared by the student to ensure they meet ethical standards and the Module Ethics Framework .	☐
c. I have completed the University's Research Integrity Training within the last three (3) years and believe I have the necessary skills and knowledge to review the documents referenced above and to offer appropriate supervision in matters of ethics.	☐
d. I am familiar with the Ethics Policy for Human Research , Research Integrity and Governance Code of Practice , Framework for the ethical review of research using Secondary Data or Publicly Available information , and Academic Integrity Policy and I will provide support to the student in following all standards relevant to their project.	☐
e. I will encourage the student to discuss with me, and reflect on, any ethical issues that arise during the project and, where relevant, I will ensure such issues are notified to the Module Leader.	☐
f. I have read the risk assessment and have considered this for the project.	☐
Signed:	
Print name:	
Date:	

Appendix C (Prototype Design & System Architecture)

The Empirisys Competitive Intelligence Engine (CIE) was delivered as a working full-stack prototype rather than a specification. The screenshots below capture the live interface; the technical description that follows documents the stack, topology and architecture that support it.

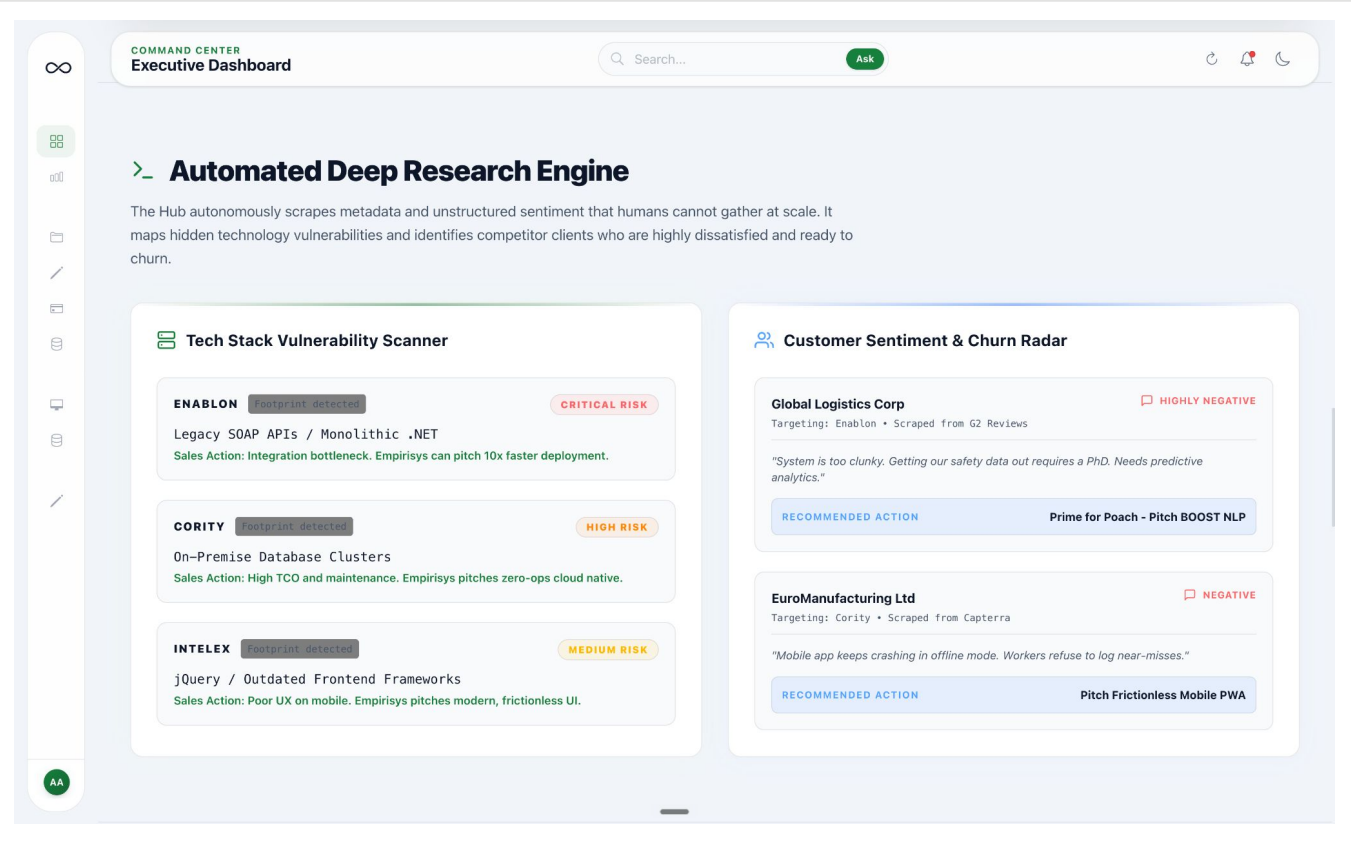
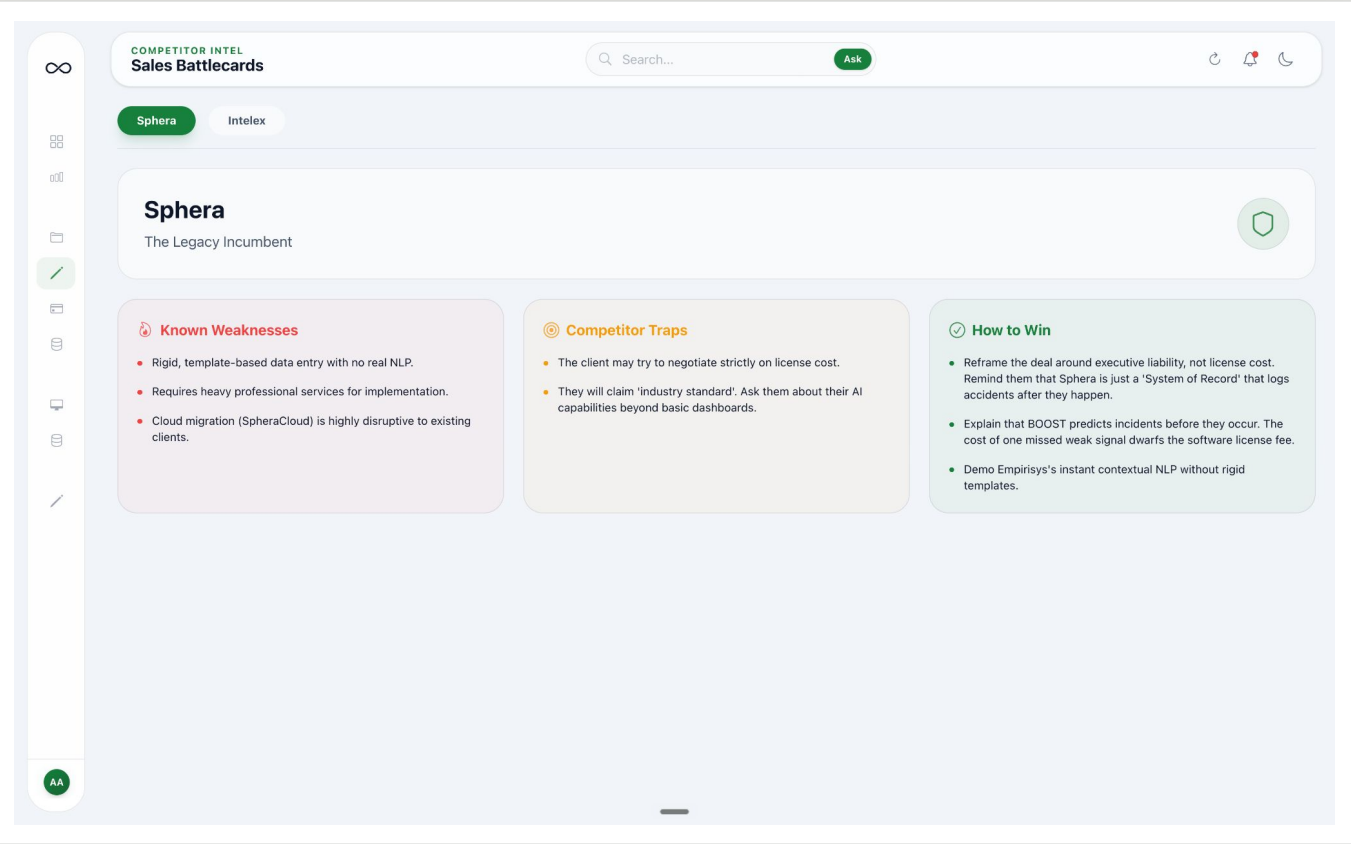


Figure C.1: High-resolution captures of the Intelligence Engine prototype

Technical Overview

C.1 TECHNOLOGY STACK

The prototype is a full-stack web application built on Next.js (App Router), combining React server and client components with TypeScript throughout. The interface is composed with Tailwind CSS, which enforces a single, token-driven design system across every view. The application is deployed as three discrete services - a frontend web app, a backend web app and a managed PostgreSQL flexible server - a topology chosen in preference to an all-in-one backend-as-a-service platform in order to retain full control over security, governance and horizontal scaling.

C.2 LAYERED TOPOLOGY

The system is organised into four cleanly separated layers. Because each layer communicates only with its immediate neighbour, the attack surface is minimised and any single tier can be scaled or replaced without disturbing the others.

THE FOUR-LAYER TOPOLOGY	
Presentation	Next.js, React and Tailwind CSS render the interface and hold no secrets.
Application / API	Server-side route handlers broker every request, enforce validation and hold all credentials.
Intelligence	A governed retrieval-augmented-generation service wrapping Anthropic's Claude API.
Data	PostgreSQL with the pgvector extension, storing structured records and vector embeddings.

C.3 SYSTEM ARCHITECTURE

End to end, a user action in the browser is dispatched to a Next.js server route, which authenticates the request, retrieves the relevant competitive intelligence from the data layer, composes a governed prompt and calls Claude, then returns a grounded, cited response to the interface. Rendering is server-first, so the client receives fast-painting pages while all privileged logic remains on the server.

Appendix D (The Data Model & Intelligence Pipelines)

D.1 PGVECTOR DATABASE SCHEMA

Competitive intelligence is stored in a normalised PostgreSQL schema extended with pgvector. A competitors table holds one row per tracked organisation, a documents table records each ingested source with its provenance and fetch timestamp, and a chunks table stores the segmented passages of each document alongside a vector embedding column. An approximate-nearest-neighbour HNSW index over the embedding column keeps semantic similarity search fast as the corpus grows, while conventional B-tree indexes support exact filtering by competitor, source type and date.

TABLE	KEY COLUMNS	PURPOSE
competitors	id, name, category, threat_level	One row per tracked organisation.
documents	id, competitor_id, source_type, url, fetched_at	Provenance of each ingested source.
chunks	id, document_id, content, embedding vector(1536)	Segmented passages with their embeddings.
Indexes	HNSW (embedding), B-tree (competitor_id, date)	Fast semantic search and exact filtering.

D.2 SECURE SERVER-SIDE API ROUTES

All privileged operations run through Next.js server route handlers. The Anthropic API key and the database connection string are read from server-only environment variables and are never shipped to the browser. Every route validates and type-checks its input, scopes database queries to the authenticated user, and applies rate limiting so that no client can exhaust the model quota. This server-side boundary is what allows the prototype to call a powerful generative model without ever exposing a credential or an unfiltered query to the client.

D.3 FIVE-STAGE CHEERIO WEB-SCRAPING PIPELINE

Public competitor sources are ingested through a deterministic five-stage pipeline built on the Cheerio HTML parser, run server-side and rate-limited to remain a polite, low-risk collector.

#	STAGE	WHAT IT DOES
1	Fetch	Retrieve the target competitor page over HTTPS with a polite user agent and timeout.
2	Parse	Load the returned HTML into Cheerio for jQuery-style server-side traversal.
3	Extract & clean	Select meaningful content nodes, strip navigation, scripts and boilerplate, normalise whitespace.
4	Chunk	Split the cleaned text into overlapping passages sized for embedding.
5	Embed & store	Generate a vector embedding for each chunk and persist it, with its metadata, into pgvector.

D.4 CLAUDE RETRIEVAL-AUGMENTED-GENERATION FLOW

A user query is first embedded and used to run a vector-similarity search against the pgvector store, returning the most relevant competitor passages. These retrieved chunks are assembled, with their citations, into a governed system prompt that fixes Empirisys’s positioning and tone. Claude then generates an answer strictly grounded in the supplied evidence, and the response is returned to the interface with its supporting sources attached, so every claim can be traced back to a specific document.

Appendix E (Client Briefing Summary & Operational Context)

The following summarises the key operational insights gathered during the project onboarding phase with the Empirisys leadership team. These briefings informed the problem diagnosis, the competitive landscape analysis, and the design of the Intelligence Engine prototype.

Peter Sueref (Co-Founder & CTO)

Peter provided an overview of Empirisys as a company: a UK-based SaaS provider operating in the Environmental Health and Safety (EHS) sector. The company's core product, BOOST, is positioned as a predictive System of Intelligence for workplace safety. The company currently operates primarily in the UK market, with strategic expansion targets across Europe and the UAE.

Andrea Rossi (Product Manager)

Andrea walked through BOOST's three-tier capability stack in detail. Tier 1 (Reactive) handles incidents already recorded. Tier 2 (Predictive) captures leading indicators such as near-misses, unsafe conditions, and overdue actions. Tier 3 (Prescriptive) applies NLP topic modelling and incident-severity frameworks to surface risk patterns before they escalate. Andrea emphasised that this predictive capability is what differentiates BOOST from legacy competitors who simply digitise paper-based filing systems.

Joe Alner (Head of Sales)

Joe outlined the competitive landscape and the core commercial challenge. Empirisys competes against established incumbents including DNV (SafeGuard), Sphera, FENNEX, and Enablon, as well as customers' own in-house DIY AI tools. The primary sales difficulty is that prospective clients perceive all EHS platforms as functionally equivalent because legacy vendors have begun retrofitting AI marketing language onto their outdated databases. Joe identified the UK as the current stronghold, with planned expansion into European markets and the UAE. The critical sales challenge is proving to a liable executive (typically a Head of Safety or Operations Director) that BOOST offers documented, auditable assurance that their organisation is prioritising the right risks, something a legacy filing system fundamentally cannot provide.

David Riskus (Senior Full-Stack Engineer)

David provided the technical briefing for the prototype build. He recommended a Next.js full-stack architecture deployed across three services: a frontend web app, a backend web app, and a PostgreSQL flexible server. He advised against Supabase for production due to limited control over security and scalability, recommending PostgreSQL directly for maximum governance. David confirmed that the prototype would integrate with Anthropic's Claude API for the intelligence layer and outlined the use of pgvector for embedding-based document retrieval. He also confirmed that post-submission, Empirisys would provision Azure resource groups for production deployment.

Appendix F (AI Platform & Agentic Workflow Overview)

F.1 PLATFORM DESCRIPTION

Beyond the single-call retrieval-augmented-generation flow described in Appendix D, the Competitive Intelligence Engine (CIE) is coordinated by a lightweight agentic layer. Rather than issuing one monolithic prompt, an incoming query is decomposed and routed to specialised agents that reason over different evidence sources in parallel, and their outputs are reconciled into a single, governed answer. This division of labour keeps each agent’s prompt narrow and auditable, and allows the platform to combine live, time-sensitive signals with the deep historical corpus held in pgvector without either source degrading the other.

F.2 AGENT ARCHITECTURE: ONE ORCHESTRATOR, THREE SUB-AGENTS

The platform runs four agents in total: a single Orchestrator that owns the workflow, and three sub-agents it dispatches to and reconciles answers from.

AGENT	TYPE	ROLE & RESPONSIBILITY
Orchestrator	Coordinator	Classifies the incoming query, decides which sub-agents are relevant, dispatches them in parallel, enforces the governed system prompt across all branches, and passes their outputs to the Synthesiser.
Live Threat Analyser	Sub-agent	Runs the Cheerio scraping pipeline (Appendix D.3) against live competitor sources on demand, surfacing pricing changes, product announcements or messaging shifts not yet present in the stored corpus.
RAG Specialist	Sub-agent	Embeds the query and runs vector-similarity search over the pgvector store (Appendix D.1), retrieving the most relevant historical chunks and their source citations.
Synthesiser	Sub-agent	Reconciles the Live Threat Analyser’s and RAG Specialist’s outputs into one Claude-generated answer, strictly grounded in the retrieved evidence, with every claim attributed to its source.

F.3 HIGH-LEVEL OPERATIONAL FLOW

#	STAGE	WHAT HAPPENS
1	Query received	A sales engineer submits a natural-language question through the Hub interface.
2	Orchestrator classifies	The Orchestrator determines whether the query needs live data, historical data, or both, and dispatches accordingly.
3	Parallel dispatch	The Live Threat Analyser and RAG Specialist run concurrently, minimising response latency.
4	Evidence returned	Both sub-agents return structured, cited evidence to the Orchestrator.
5	Synthesis	The Synthesiser composes one grounded answer from the combined evidence via the Claude API.
6	Governed delivery	The response, with its citations, is returned to the sales engineer in the Hub interface.

F.4 SUMMARY FOR A BUSINESS LEADER

In plain terms, the platform works like a small research team rather than a single search box. One team lead (the Orchestrator) reads the question and decides who on the team needs to look into it: one specialist checks what a competitor is doing right now, another pulls up everything Empirisys already knows from past cases, and a third writes up a single, clear answer that references exactly where each fact came from. The result is an answer a sales engineer can trust and defend in front of a client, because it is never a guess, it is always traceable back to a real, cited piece of evidence.

F.5 AGENTIC ARCHITECTURE - VISUAL REPRESENTATION

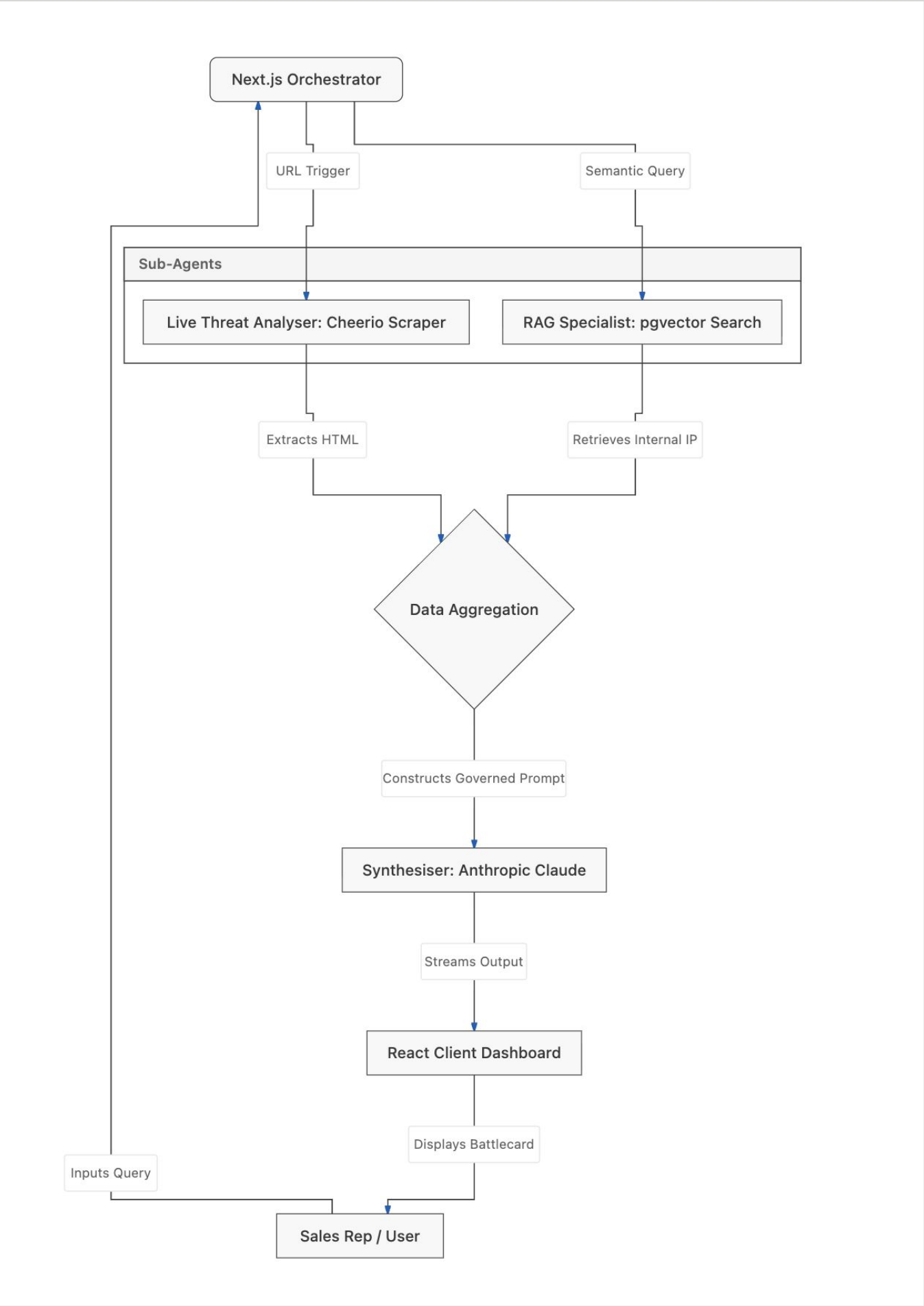


Figure F.1: The four-agent orchestration model - one Orchestrator dispatching two parallel sub-agents into a single Synthesiser.